



**SCHOOL OF ELECTRICAL ENGINEERING
AND TELECOMMUNICATIONS**

Design of a Multi-Channel MPPT System for Solar Vehicle Applications

Executive Summary

by

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Thesis submitted as a requirement for the degree
Bachelor of Engineering (Electrical Engineering)

Submitted: August 3, 2026
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1 Justification for Thesis

Sunswift Racing is an ambitious student project focused on pushing the boundaries of solar car technology. The team has now transitioned to developing their next vehicle, Sunswift 8. This is a *TriBrid* ADR compliant vehicle that contains three distinct power sources: battery, hydrogen, and solar power.

Integrating solar arrays into an automobile poses a set of engineering constraints that differ greatly from conventional solar projects. These include, but are not limited to: large differences in effective irradiance between arrays, frequent partial shading, varying temperature gradients, tight space constraints, and harsh automotive environments. SR8 carries five arrays across the front, roof and rear body panels, and because these panels are curved the effective angle between arrays can be as large as 20° . This alone represents roughly a 6% difference in received irradiance before any shading is considered. Traditionally, solar arrays are connected in series or parallel and a single MPPT is used to extract power, which forces every array onto one operating point and limits the whole system to its weakest string. On a vehicle generating 769 W at peak, the resulting mismatch losses are significant.

Commercially available MPPTs are designed for static installations and are packaged as single-channel modules. Building a five-channel system from these parts would require five separate units, each with its own enclosure, mounting provisions and harness, in a vehicle where space and mass are tightly constrained. These units also do not natively report into the SR8 telemetry system. A purpose-built module that places all five tracking channels on a single board removes this duplication, allows the protection and FD-CAN telemetry required by the vehicle to be designed in from the start, and gives the team full control over the tracking behaviour under the dynamic conditions a car experiences.

The goal of this thesis is therefore to develop an integrable product for Sunswift 8 that extracts maximal energy from the onboard photovoltaic arrays and delivers it to the auxiliary power systems in the vehicle. This thesis proposes per-string power-point tracking using GaN synchronous boost converters, with all MPPT modules integrated on a single board. Beyond SR8, the same problem applies to any production vehicle carrying body-mounted PV, where curved surfaces and frequent shading are unavoidable, and the design will carry directly into future Sunswift vehicles.

2 Objectives

This thesis is focused on the design of a multi-channel MPPT module for automobiles with integrated PV arrays, allowing maximal efficiency in the highly dynamic environmental conditions that affect PV

arrays on vehicles. This module will be developed in conjunction with Sunswift Racing, with design decisions focused on optimising power extraction from SR8's onboard PV arrays.

The ultimate objective is to deliver Sunswift Racing a robust, integrable product that requires very little configuration to achieve optimal performance. This product will acknowledge the unique operating conditions of a PV system in an automotive environment, and take appropriate design measures to limit or negate mismatch losses. This is broken down into the four objectives below:

1. Design and validate a per-string MPPT architecture, with an independent tracking channel for each of the five SR8 PV arrays, delivering a combined 769 W peak into the 46.8 V nominal LV bus.
2. Achieve a converter efficiency above 96% across the expected operating range of each channel, verified by measurement on the final hardware.
3. Research and implement MPP tracking algorithms suited to the rapidly changing irradiance conditions of a moving vehicle, and develop the firmware required to run them across all five channels concurrently.
4. Integrate the protection and telemetry required for safe operation within the SR8 electrical architecture, including cycle-by-cycle overcurrent limiting, reverse polarity protection, and FD-CAN reporting to the vehicle.

3 Literature Review

PV cells convert light into electrical power through the photovoltaic effect and exhibit a non-linear IV curve that is constantly moving [7, 11]. Irradiance increases I_{SC} approximately linearly while only increasing V_{OC} logarithmically [11], and temperature causes a slight rise in I_{SC} but a significant reduction in V_{OC} and output power [7]. Since effective irradiance depends on the angle of incidence, the curved and differently oriented arrays on SR8 will sit on different IV curves, which is why an MPPT is required [4].

When cells with different characteristics are interconnected, the system must operate at a single point and no cell sits at its own MPP [6]. In series strings the current is limited by the weakest cell, creating mismatch losses [9]. Bypass diodes prevent hot-spot damage but remove cells from the string and introduce multiple local maxima, which simple algorithms such as perturb and observe are likely to get trapped in [9].

Since the array voltage sits below the LV bus, a non-isolated boost topology is the appropriate choice for the DC/DC stage [3]. Replacing the diode with a second switch removes the forward-drop loss of an asynchronous design. The key trade-off is switching frequency, as raising it reduces current and voltage ripple but increases switching and gate-charge losses [3]. GaN devices have very low gate charge and switch far faster than silicon, at the cost of sensitive layout [12].

Two dominant architectures exist. An SMPPT system connects all arrays in series and/or parallel into a single tracker. This is the simplest and lowest cost option and is highly efficient under uniform conditions. Its drawback is that the whole system is forced onto one operating point [1]. As soon as one cell operates at a lower I_{MPP} the entire string is limited by it, and parallel strings are held at a single terminal voltage regardless of their individual optima [6]. SMPPT systems therefore suit static installations where shading can be planned out, rather than automotive systems with curved mounting surfaces and frequent partial shading [2].

Distributed MPPT systems instead segment the PV system across multiple modules, each with its own operating point [1]. String and module level MPPTs raise cost and integration complexity, but every additional MPPT adds flexibility under dynamic conditions and allows sections at different angles [5]. DC optimisers apply the same idea to groups of cells within a string, but require arrays without bypass diodes and access to intermediate connection points, which is likely not an option in space limited systems [8, 1]. Distributed systems are more complex to integrate but far more robust and scalable [10], making a per-string architecture the optimal approach for SR8.

4 Preliminary Work

Thesis A established the requirements for this system, including the bus voltage, panel V/I characteristics, communication interfaces, and the loss mechanisms present in synchronous switching converters. Thesis B began with a single module test board, built to characterise my chosen inductor, GaN switching device and capacitors before committing to a five-channel single-board implementation. It used a $33\mu H$ SER2211 inductor, an LMG2100R026 GaN half-bridge, and two INA228 power sensors on the input and output, driven by an STM32H743 over a HI/LI interface.

Two faults came out of this board. The first was my own error in the timer implementation, where I had assumed the high-side FET was the controlling switch, when in a synchronous boost it is the low-side FET. This caused the converter to start at 100% duty, and was rectified by repinning the JST-GH

cables. The second was more involved. The digital power sensors became noisy and dropped out as current increased, worsening with switching frequency and power throughput, which pointed to an EMI related issue. I first found that the I2C bus had been routed under the edge of the inductor because the footprint keepout did not extend far enough, and rerouting it helped but did not solve the problem. Measuring the inductor ripple with my supervisor showed $\Delta I \approx 4A$ at a nominal $8.22A$ input, saturating the inductor every cycle, but moving the flux leakage path away from the sensors changed nothing. The root cause was the return current path, seen in figure 1. I had not stitched the ground plane surrounding the INA sensors, forcing the return current through the SW node of the FET. Cutting the plane and rewiring the sensors directly solved the dropouts entirely.

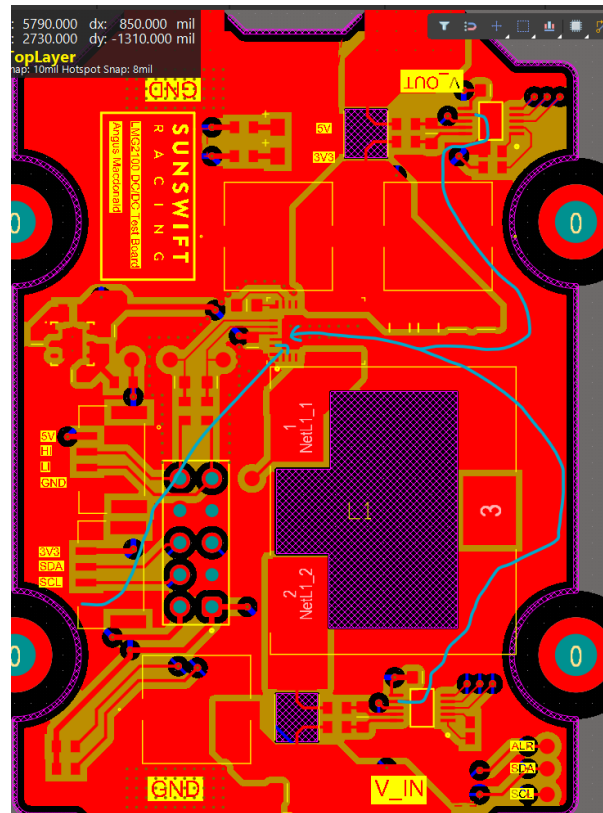


Figure 1: Return current path illustrated on layout

Whilst this was being resolved I collected power stage data manually using a bench supply and load with voltage sense wires. At $300kHz$ peak efficiency was approximately 98%, shown in figure 2, and a frequency sweep at a constant $7A$ gave 98.4% at $200kHz$, although the ripple at this frequency is likely unacceptable for accurate tracking. Both exceed my Thesis A prediction of 96.5%, which I attribute to shorter rise and fall times on the FET's than I had assumed, though measurement error in the supply, load and shunt means these should be treated as estimates.

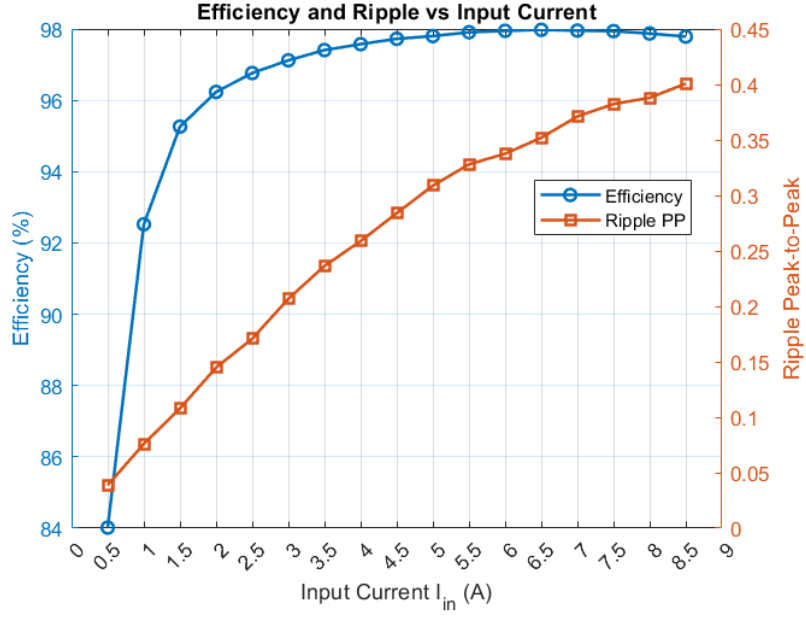


Figure 2: Efficiency and ripple against input current at $f_{sw} = 300kHz$

The remainder of Thesis B was spent on the final implementation, shown in figure 3. The most significant addition is hardware cycle-by-cycle current control and overcurrent limiting using an INA310A2, programmed to trip at 12A, since the $100\mu s$ update rate of the INA228 would allow a spike of around 82A to develop between detection points. I also moved to a VER2923 inductor with more saturation headroom, added an ideal diode on the auxiliary bus output with a bypass switch so the solar team can inject current into the panels for EL imaging, and implemented the microcontroller supplies alongside the FD-CAN, USB and programming interfaces.

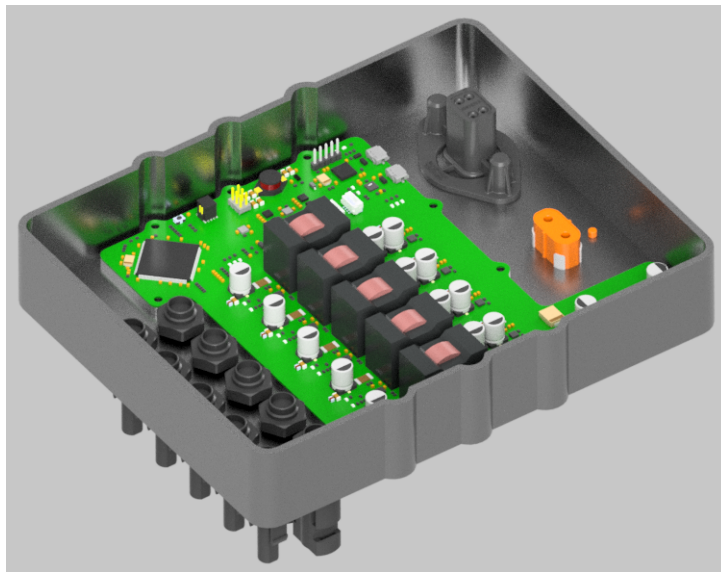


Figure 3: 3D view of current design

Since the Thesis B report, layout and routing have been completed on a 6-layer stackup, chosen in response to the EMI problems faced on the test board and the mixed analog/digital design. I have also begun the firmware, starting with the peripheral drivers and a hardware abstraction layer, so that the tracking and control code can be developed independently of the underlying hardware.

5 Plans

The final board has now been released for manufacture, and I expect it to return assembled in about two weeks. This puts me roughly a month behind my original plan, mostly due to the time spent diagnosing the EMI issues on the test module. While the board is being manufactured I will continue with the firmware, building on the drivers and hardware abstraction layer I have started, so that bring-up can begin as soon as the hardware arrives. Characterisation of the boost stages and the auxiliary circuits will follow, and the control loops and MPPT algorithms will take up most of the break and the early weeks of term 3. The original scope also included an isolated converter for charging the vehicle's high voltage battery. This has since been de-scoped and will be replaced with a commercial part, allowing the work to concentrate on the LV tracking system. The remainder of thesis C will be spent on integration testing, designing the poster, and writing the thesis.

6 Meeting Log

Meeting Date	Discussion points
Week 2, Friday	Initial schematic reviews
Week 3, Friday	Board debugging, removal of sensors, inductor rotation, and discussing reasons for the issues
Week 4, Friday	Discussion of cycle-by-cycle current limiting and control, and how to implement it
Week 6, Friday	Detailed schematic review and initial layout reviews
Week 7, Friday	Full schematic and layout reviews

Table 1: Meeting log

References

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